

Spectral Theory by the Lakes

Lancaster University, 9-10 April, 2026

Program & Abstracts

Thursday 09.04.2026

09'00 - 09'30 Registration (Charles Carter Foyer)

- Charles Carter A15 -

09'30 - 10'30	<i>Eugene Shargorodsky</i> : Estimate for the Morse index of a Stokes wave
10'30 - 11'00	<i>Sukrid Petpradittha</i> : Lieb–Thirring type inequalities for Schrödinger operators with complex potentials

11'00 - 11'30 Coffee Break (Charles Carter Foyer)

- Charles Carter A15 -

11'30 - 12'30	<i>Mira Shamis</i> : TBA
---------------	--------------------------

12'30 - 14'00 Lunch Break

- Charles Carter A15 -

14'00 - 15'00	<i>Alexander Pushnitskii</i> : Periodic Hankel operators
15'10 - 16'10	<i>Leonid Pastur</i> : Ergodic Hankel operators

16'10 - 16'30 Coffee Break (Charles Carter Foyer)

- Charles Carter A15 -

16'30 - 17'30	<i>Jakob Reiffenstein</i> : TBA
---------------	---------------------------------

Friday 10.04.2026

- Charles Carter A15-

09'30 - 10'30	<i>Catherine Drysdale:</i> TBA
10'30 - 11'00	<i>Benedikt Buchecker:</i> TBA

11'00 - 11'30 Coffee Break (Charles Carter Foyer)

- Charles Carter A15 -

11'30 - 12'30	<i>Leonid Parnowski:</i> TBA
---------------	------------------------------

12'30 - 14'00 Lunch Break

- Charles Carter A15 -

14'00 - 15'00	<i>Yan-Long Fang:</i> TBA
---------------	---------------------------

15'00 - 15'30 Coffee Break (Charles Carter Foyer)

- Charles Carter A15 -

15'30 - 16'30	<i>Noema Nicolussi:</i> TBA
---------------	-----------------------------

Spectral Theory by the Lakes 2026 – Abstracts

title

Buchecker Benedikt

date

abstract

title

Drysdale Catherine

date

abstract

Scattering resonances of transmission problems

Fang Yan-Long

date

Understanding wave propagation in the presence of an obstacle requires detailed knowledge of the associated scattering resonances. In this talk, I will outline how the geometry of the obstacle is reflected in the distribution of these resonances. Beyond geometric effects, the refractive index of the obstacle also plays a crucial role in shaping the resonance structure. In particular, for a certain range of negative refractive indices, a new class of resonances emerges, which are known as surface plasmons. I will discuss key properties of these plasmons, along with a Weyl-type asymptotic formula that describes their distribution. This is based on a joint work with Jeffrey Ga

title

Nicolussi Noema

date

abstract

title

Parnowski Leonid

date

abstract

Ergodic Hankel operators

Pastur Leonid

date

Abstract. We introduce a new class of operators: ergodic families of selfadjoint Hankel operators. Inspired by the spectral theory of differential and finite-difference operators with ergodic coefficients, we develop a spectral theory of this class. We define the Integrated Density of States (IDS) measure and establish its fundamental properties. In particular, we determine the total mass of the IDS measure in the positive semi-definite case. We also consider in more detail the subclass of ergodic Hankel operators which we call the random Kronig–Penney–Hankel (rKPH) model. We prove the counterparts of the cornerstone results of the spectral theory of random Schrödinger operators: Lifshitz tails at the edges of the spectrum, the Wegner estimates for the IDS, and the Anderson localisation in a natural asymptotic regime. This is joint work with Alexander Pushnitsky (King’s College London).

Lieb–Thirring type inequalities for Schrödinger operators with complex potentials

Petpradittha Sukrid

date

In this talk, I will present recent developments in Lieb–Thirring (LT) type inequalities for non-self-adjoint Schrödinger operators. In view of joint work (2025) with S. Bögli and F. Štampach, we show that the spectral bound concerning a possible generalization of the LT inequality, announced by Demuth, Hansmann, and Katriel in the Integral Equations and Operator Theory in 2013, is false. Then, in recent joint work (2026) with S. Bögli, we prove new LT type inequalities by means of a new eigenvalue counting estimate for non-self-adjoint operators.

Periodic Hankel operators

Pushnitskii Alexander

date

I will discuss spectral properties of a class of self-adjoint Hankel operators H , realised as integral operators on the positive semi-axis. Operators of this class (we call them periodic Hankel operators) are those that commute with dilations by a fixed factor. In analogy with the spectral theory of periodic Schroedinger operators, periodic Hankel operators H admit the Floquet-Bloch decomposition, which represents H as a direct integral of certain fiber operators with discrete spectra. As a consequence, operators H have band spectra (the spectrum of H is the union of disjoint intervals). A striking feature of this model is that flat bands (i.e. intervals degenerating into points, which are eigenvalues of infinite multiplicity) are possible; I will discuss some simple explicit examples of this nature. The spectral analysis of periodic Hankel operators is based on the theory of elliptic functions; if time permits, I will explain this connection. This is joint work with Alexander Sobolev (University College London).

title

Reiffenstein Jakob

date

abstract

title

Shamis Mira

date

abstract

Estimate for the Morse index of a Stokes wave

Shargorodsky Eugene

date

Stokes waves are steady periodic water waves on the free surface of an infinitely deep irrotational two-dimensional flow under gravity without surface tension. They can be described in terms of critical points of a certain functional; this allows one to define the Morse index of a Stokes wave. I will discuss some known and new estimates for the Morse index of a Stokes wave.
